Author Name : Jamaalullah Abdul Rahman

Student ID : 844455779

Estimated time to complete project : 35-40 hours

1 Free body diagram and calculation

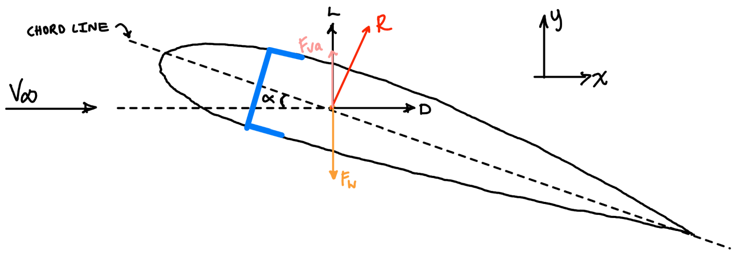

Figure 1: Free body diagram of the Aerofoil

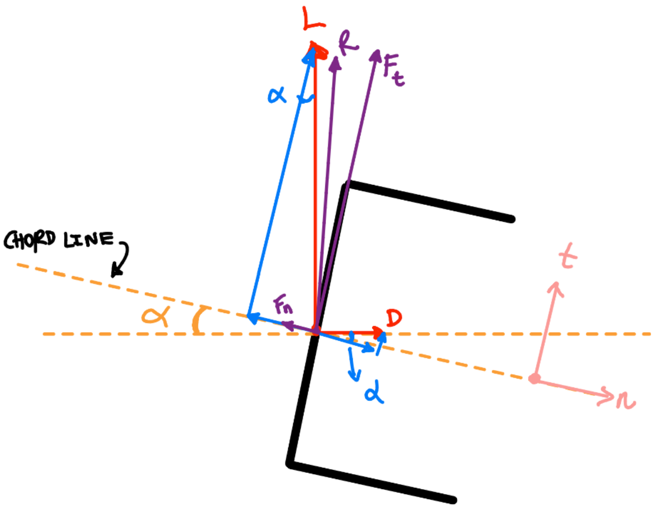

Figure 2: Free body diagram of forces acting on C bar

FORCE CALCULATIONS:

$$
\begin{aligned}
\Sigma\text{Forces in Y direction} = F_{Lift} - F_{weight} &= ma_y \\
F_{Lift} &= F_{Vertical\ acceleration} + F_{weight} \\
F_{Lift} &= [92.5kg \times (2.9 \times 9.81)] + (92.5 \times 9.81) \\
&= 3539\ N \\
\Sigma\text{Forces in X direction} &= F_{Drag} \\
F_{Drag} &= \frac{F_{Lift}}{85} \\
&= 41.6\ N
\end{aligned}
$$

$$
\begin{aligned}
\Sigma\text{Forces in n} &= F_{Drag} cos(\alpha) - F_{Lift} sin(\alpha) \\
&= 41.6 \cdot cos(3) - 3539 \cdot sin(3) \\
&= -143.64\ N \\
\Sigma\text{Forces in t} &= F_{Drag} sin(\alpha) + F_{Lift} cos(\alpha) \\
&= 41.6 \cdot sin(3) + 3539 \cdot cos(3) \\
&= 3536.3\ N
\end{aligned}
$$

CHECK RESULTANT

$$
\begin{aligned}
Resultant &= \sqrt{\text{Lift}^2 + \text{Drag}^2} \\
&= \sqrt{3539^2 + 41.6^2} \\
&= 3539.2\ N \\
Resultant &= \sqrt{F_t^2 + F_n^2} \\
&= \sqrt{3536.3^2 + (-143.64)^2} \\
&= 3539.2\ N
\end{aligned}
$$

Resultant of lift and drag forces equals to resultant of tangential and normal forces. Therefore, the forces acting on the beam is 3536.3N and -143.64N.

2 CAD model and FEA geometry

2.1 Shell Model

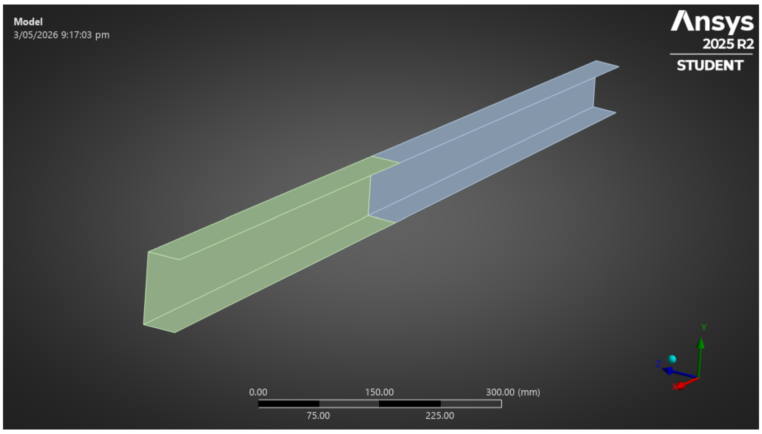

Figure 3: Shell model in ANSYS mechanical

2.2 Solid model

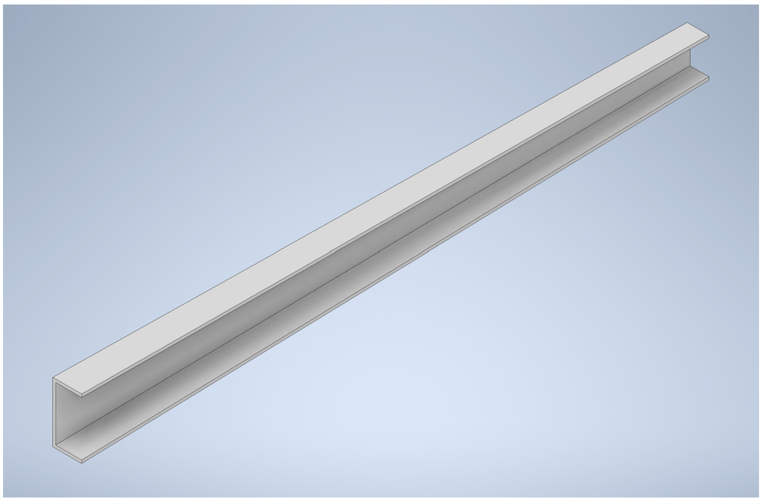

Figure 4: 3D solid model in Inventor

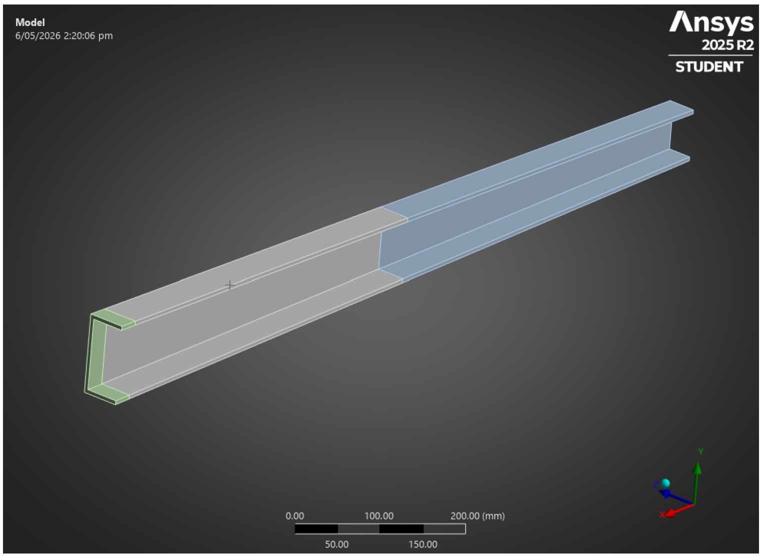

Figure 5: Solid model in ANSYS Mechanical

3 Load Application

3.1 Centroid Calculations

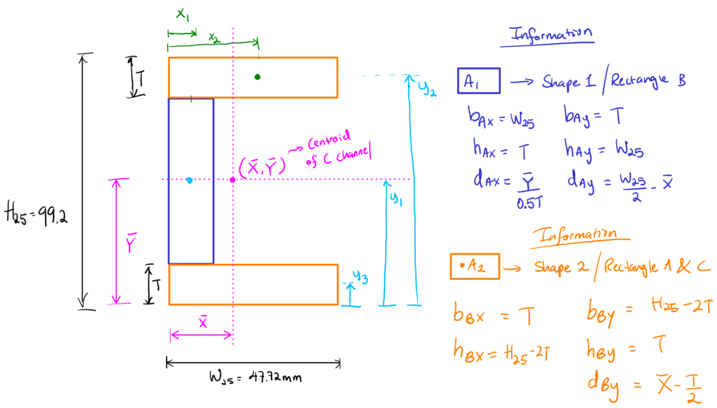

Figure 6: Centroid calculation diagram

To calculate the centroid, the dimensions of the C channel 25mm away from the fuselage had to be calculated.

$$H_{25} = \frac{H_1 + (H_2 - H_1) \cdot 25}{L} = 99.17\ mm$$
$$W_{25} = \frac{W_1 + (W_2 - W_1) \cdot 25}{L} = 47.72\ mm$$

To calculate the x and y coordinates of the C channel, the bar had to be split and the individual area and centre distances were calculated.

$$\bar{Y} = \frac{A_1 \cdot y_1 + A_2 \cdot y_2 + A_2 \cdot y_3}{A1 + A2 + A3} = 49.6\ mm$$
$$\bar{X} = \frac{A_1 \cdot x_1 + 2 \cdot (A_2 \cdot x_2)}{A1 + A2 + A3} = 16.12\ mm$$

3.2 FEA remote forces

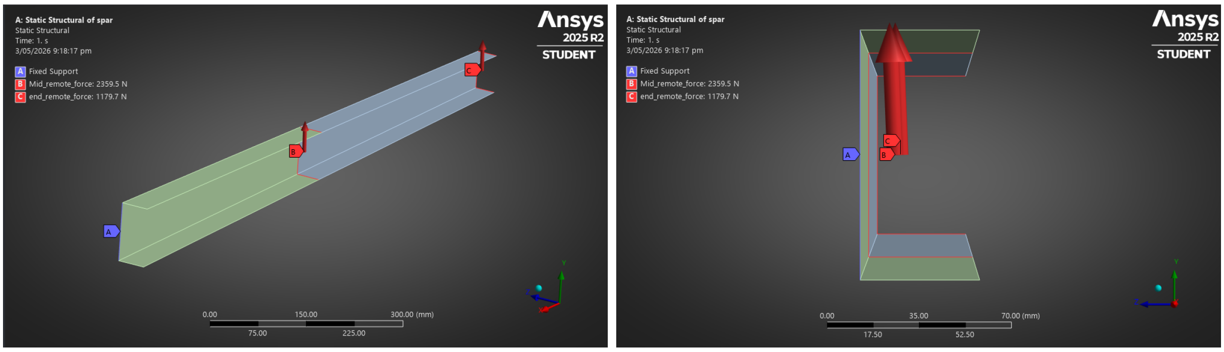

Figure 7: Remote forces acting on shell model

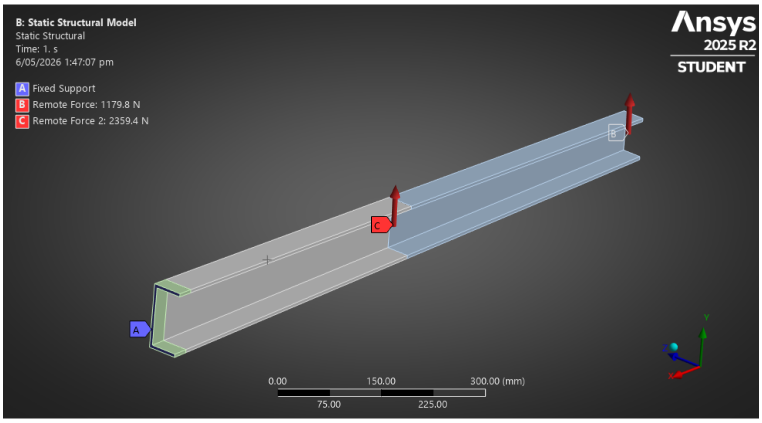

Figure 8: Remote forces acting on 3D model

4 Support application

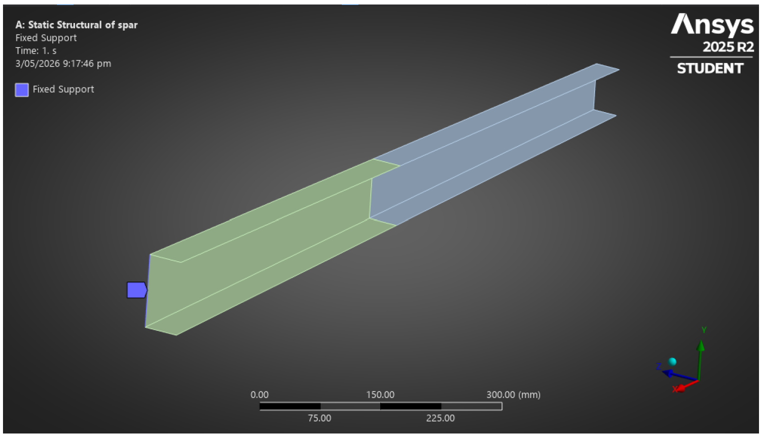

Figure 9: Support application on shell model

5 Mesh convergence study - shell model

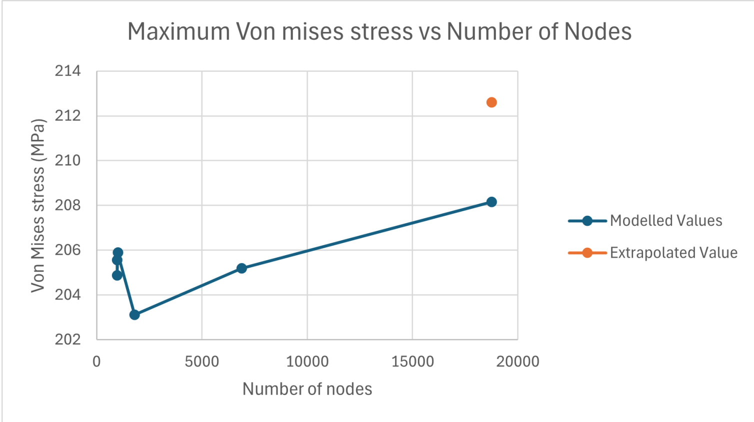

Figure 10: Von-mises stress shell

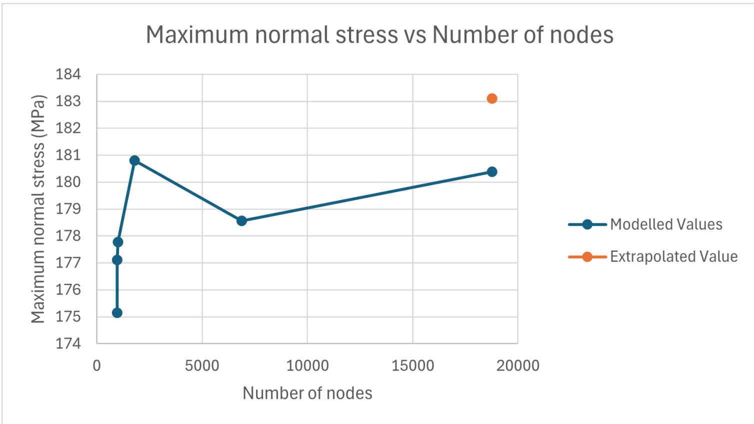

Figure 11: Maximum normal stress shell

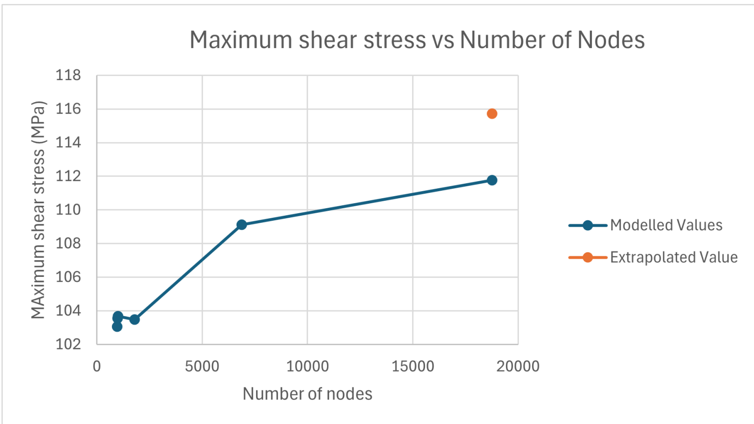

Figure 12: Maximum shear stress shell

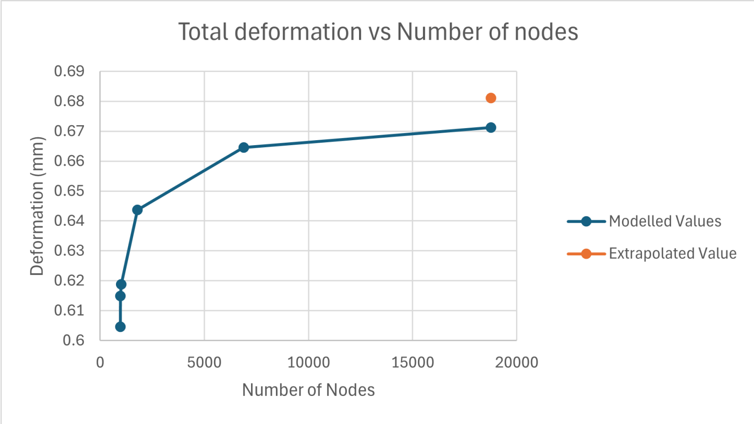

Figure 13: Total deformation shell

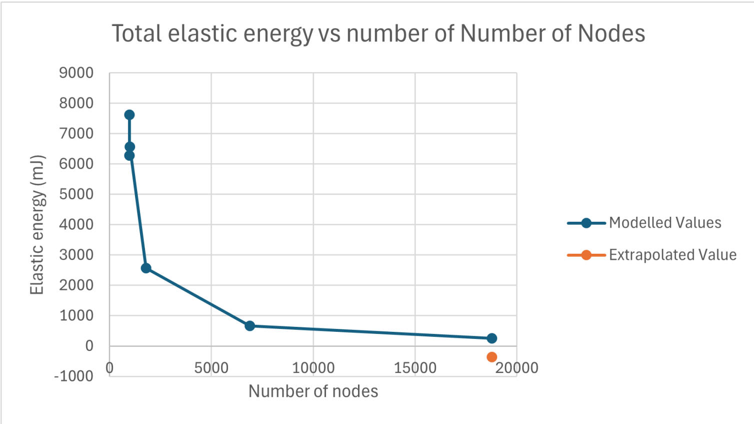

Figure 14: Total elastic energy shell

Precentage error against the extrapolated value								
r_12	1.67	extrapolated soln	Mesh element size(m)	Solution time(s)	Error for von mises %	Error for max shear stress %	Error for max normal %	Error for deformation %
	Von mises	212.610	30	20	3.313	10.951	4.347	11.246
	Max Shear stress	115.728	20	23	3.636	10.512	3.272	9.724
	Max normal	183.110	15	18	3.156	10.411	2.909	9.157
	Displacement	0.681	10	20	4.465	10.582	1.264	5.501
			5	22	3.491	5.705	2.484	2.435
			3	42	2.094	3.423	1.491	1.461

Figure 15: Shell information

6 Mesh convergence study - solid model

6.1 Tetrahedral mesh

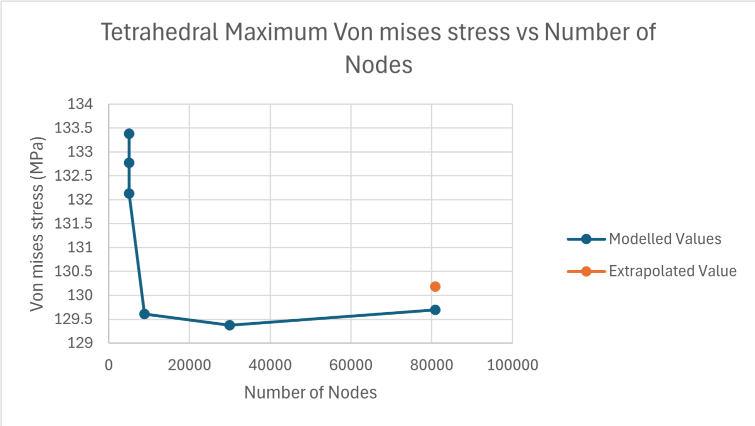

Figure 16: Von-mises stress tetrahedral

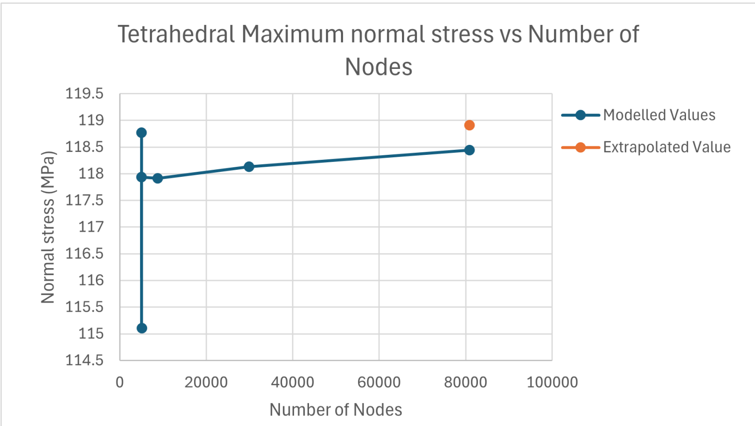

Figure 17: Maximum normal stress tetrahedral

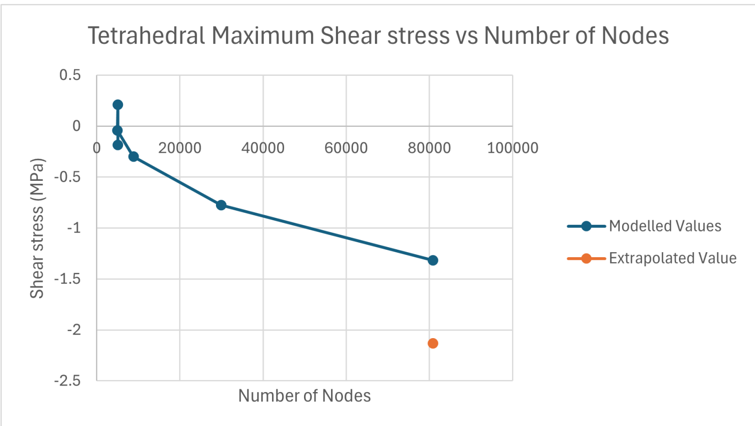

Figure 18: Maximum shear stress tetrahedral

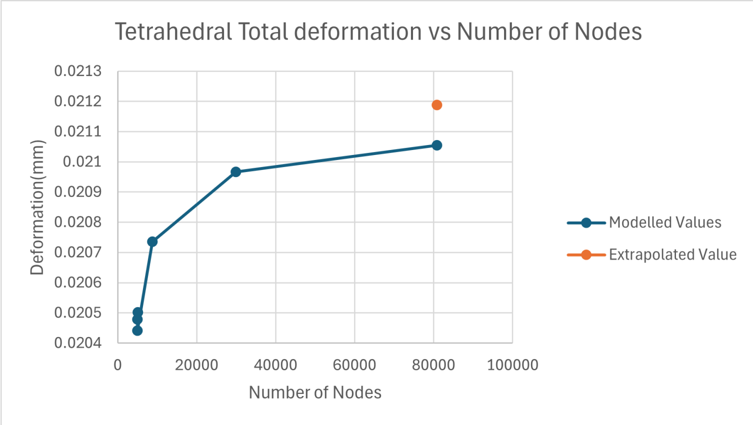

Figure 19: Total deformation tetrahedral

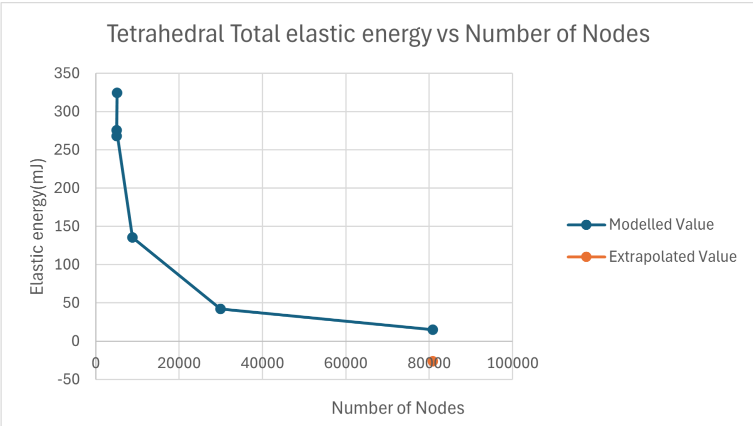

Figure 20: Total elastic energy tetrahedral

6.2 Hex mesh

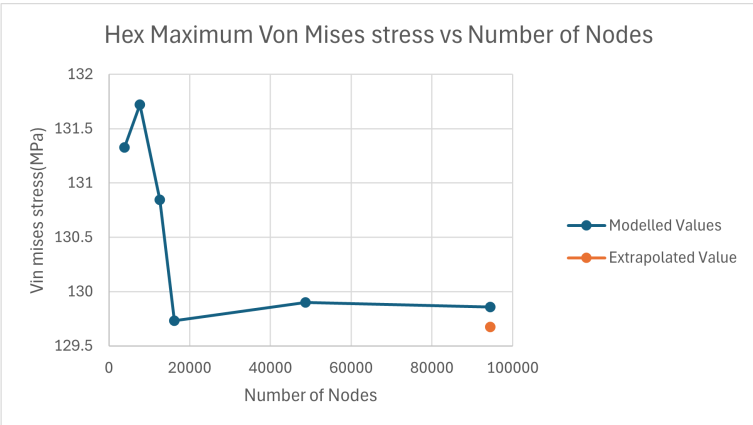

Figure 21: Von-mises stress hex

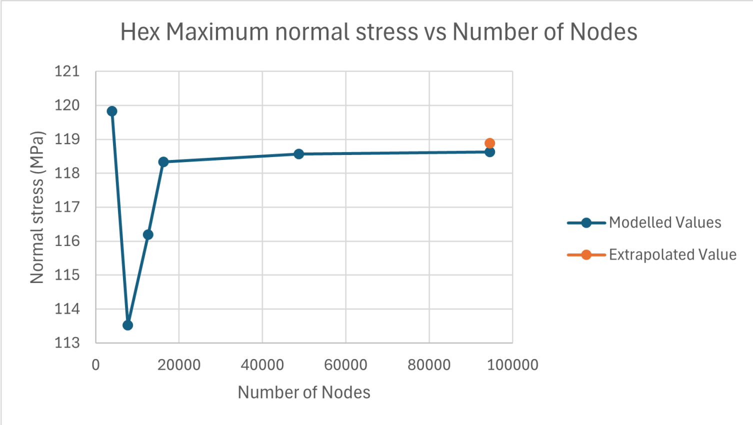

Figure 22: Maximum normal stress hex

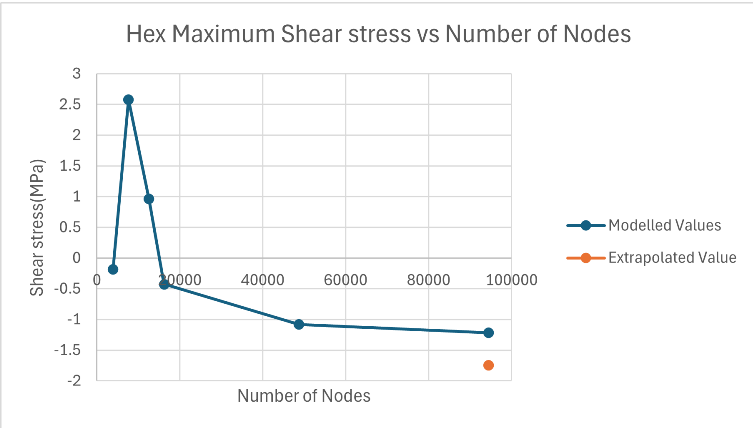

Figure 23: Maximum shear stress hex

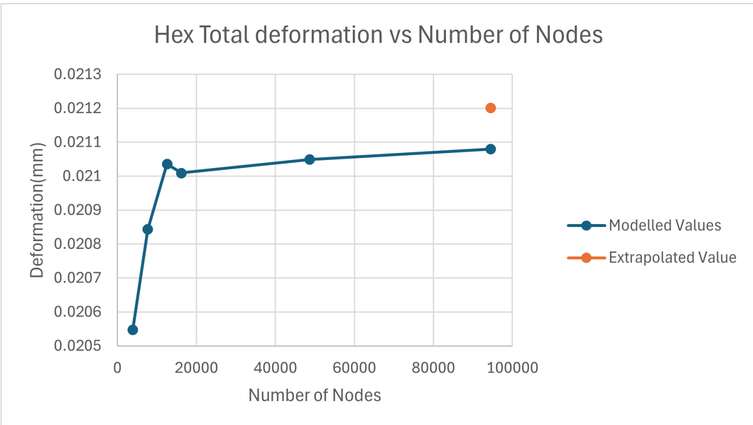

Figure 24: Total deformation hex

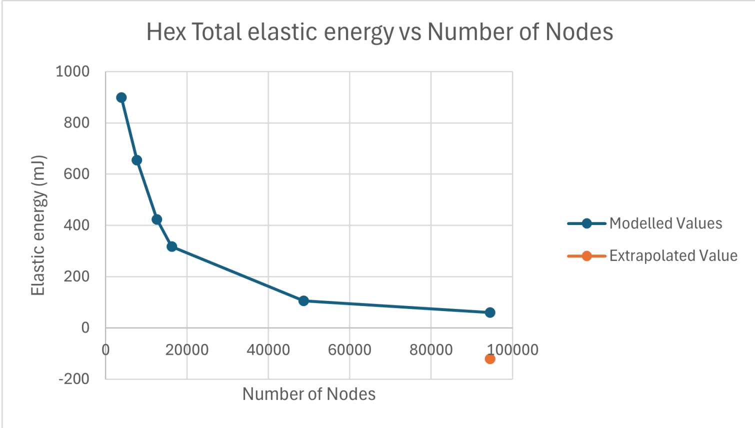

Figure 25: Total elastic energy hex

				Tet Error %		
r_12 = 1.67	Extrapolated value	Mesh element size(mm)	Solution time (s)	Error for Von mises	Error for deformation	Error for normal stress
von mises	130.18	30	20	2.458	3.350	0.119
deform	0.02	20	20	1.993	3.238	3.200
normal	118.91	15	20	1.498	3.526	0.818
		10	21	0.442	2.136	0.840
		5	33	0.619	1.044	0.656
		3	59			
				Hex Error %		
von mises	129.68	30	20	1.274	3.410	0.097
deform	0.02	20	20	1.577	3.298	3.179
normal	118.89	15	22	0.902	3.586	0.796
r_12 = 1.25		10	22	0.042	2.197	0.818
		5	32	0.174	1.105	0.634
		4	50	0.140	0.688	0.371

Figure 26: 3D model information

7 Whole-of-life embodied carbon material selection

With a global mesh element size of 3mm, the following data was collected using the 2D shell model.

Varying thickness	Aluminium mass(kg)	aluminium Von mises (3mm)	Steel mass (kg)	Steel Von mises (3mm)	ECO2 Aluminium (kg)	ECO2 Aluminium (kg)
6	2.74	176.36	7.78	177.57	44.91	375.62
5.5	2.52	192.25	7.13	193.55	38.44	309.62
5	2.29	211.13	6.48	212.54	32.66	250.61
4.5	2.06	234.02	5.83	235.57	27.54	198.43
4	1.83	262.48	5.18	264.21	23.09	152.92
3.5	1.60	298.96	4.54	300.91	19.26	113.91
3	1.37	347.49	3.89	349.73	16.06	81.18
2.5	1.14	415.22	3.24	417.81	13.44	54.54
2	0.91	516.18	2.59	519.22	11.40	33.71

Figure 27: Embodied carbon of materials

$$\text{Embodied C02}_{Aluminium(5mm)} = 3.86 \times 2.29^{2.23} + 8.24 = 32.7\,kg$$

$$\text{Embodied C02}_{Steel(5mm)} = 3.86 \times 6.48^{2.23} + 1.42 = 250.6\,kg$$

8 Validation calculations

8.1 Total mass of spar

Table 1: Table of C bar dimensions

Dimensions of C bar	H1	H2	W1	W2	L	T
Given values (mm)	100	65	48	36	1055	5

The C channel is broken down into 3 segments; Top flange, Bottom flange and Web. By calculating each segment's volume, the total volume can be found.

FLANGE CALCULATIONS

$$(W_1 \cdot L) - \frac{L \cdot (W_1 - W_2)}{2} = 44310\ mm^2$$
$$44310 \times T = 221550\ mm^3$$

WEB CALCULATIONS

Height and width of web has been reduced to take thickness of flanges into account. Therefore the new height and width of the web is 90mm and 55mm.

$$\text{New}H_1 \cdot L - \frac{L \cdot (\text{New}H_1 - \text{New}H_2)}{2} = 76488\ mm^2$$
$$76488 \times T = 382440\ mm^3$$

TOTAL VOLUME AND MASS OF SPAR

$$\begin{aligned}
\text{Volume of spar} &= (2 \times 221550) + (382440) \\
&= 825540 mm^3 \\
\text{Mass of steel spar} &= 825540 \times 7.85 \times 10^{-3} \\
&= 6480.5g \\
\text{Mass of aluminium spar} &= 825540 \times 2.7 \times 10^{-3} \\
&= 2229g
\end{aligned}$$

8.2 Maximum normal stress

To calculate validation segments, moments of area was needed. The shape was split as seen in the centroid figure, and the second moment of area of each rectangle in each axis was calculate.(See Figure 6)

$$I_{Ax} = \frac{b_{Ax} \cdot h_{Ax}^3}{12} + A_2 \cdot d_{Ax}^2 = 529430\ mm^4$$
$$I_{Ay} = \frac{b_{Ay} \cdot h_{Ay}^3}{12} + A_2 \cdot d_{Ay}^2 = 59529\ mm^4$$
$$I_{Bx} = \frac{b_{Bx} \cdot h_{Bx}^3}{12} = 295430\ mm^4$$
$$I_{By} = \frac{b_{By} \cdot h_{By}^3}{12} + A_1 \cdot d_{By}^2 = 83708\ mm^4$$

The 2nd moment of area for rectangle A and rectangle C are the same. The full 2nd moment of area can be calculated as seen below.

$$I_{xx} = 2 \times I_{Ax} + I_{Bx} = 1354300\ mm^4$$
$$I_{yy} = 2 \times I_{Ay} + I_{By} = 202770\ mm^4$$

Taking moments about the 25mm slice, there are 2 moments acting on the x and y axis.

$$M_x = F_{Vertical\ end} \times (L - 25) = 1.1788 \times 10^3\ N \times 1030mm = 1.214 \times 10^6 Nmm$$
$$M_y = F_{Horizontal\ end} \times (L - 25) = -47.88\ N \times 1030mm = -4.932 \times 10^4 Nmm$$

$$\sigma_{normal} = \frac{M_x \cdot \bar{Y}}{I_{xx}} + \frac{M_y \cdot (W_{25} - \bar{X})}{I_{yy}} = 52.1\ MPa$$

8.3 Maximum transverse shear stress

$$Q_{shear} = \frac{H_{25} - 2T}{4} \times \frac{T \cdot H_{25} - 2T}{2} = 4.97 \times 10^3$$
$$\tau_{shear} = \frac{F_{vertical\ mid} \cdot Q_{shear}}{I_{xx} \cdot T} = 1.7MPa$$

8.4 Maximum von Mises stress

$$\sigma_{Von\ Mises} = \sqrt{\sigma_{Normal}^2 + 3\tau_{shear}^2} = 52.2MPa$$

9 Discussion

1. Hand calculations helped verify that the forces were placed in the correct vicinity of the 2D shell, as the centroid aligned with the remote force location. However,the 2D model differed noticeably to the hand calculations, especially with the normal stress, likely because the hand calculations did not account for twisting. The shear stress calculated was slightly larger than the modelled stress. This could be due to the differences in a perfect theoretical scenario, to the model in ANSYS which calculated shear along the path with local mesh densities. The shell model accuracy was limited as it did not represent through

thickness stress. It could not validate the result of the forces accurately. Although convergence is seen in the finer meshes, several outliers in the ANSYS model where stresses spiked unexpectedly. This could be due to meshing issues, or stress singularities along the path used.

2. The advantage of modelling with a 2D shell was that the time taken to mesh the model was significantly lesser, the placement of the forces was much easier and easier to visualise. A shell could give a better understanding on how the forces act on the beam. The disadvantages of modelling with the 2D model was that the stress throughout the thickness was 0 as it was calculating stresses on a plane ($\sigma_z = 0$). The modelled values from a 3D model gave a more accurate representation of the whole bar. However, stress singularities were more prominent in the 3D model. This made the convergence study less robust, as there are stress singularities in the graph. This also affected the error of extrapolated points of the 3D model, especially in the shear stress and total elastic energy, which were ultimately removed.

3. In the model, the maximum stress points occurred on the corner edges of the fuselage connection (see Figure 7). By calculating the stresses along the fuselage support edge, the chances of modelling a stress singularity is high. The C channel has sharp edges, which would result in an indefinite stress at those corner points. By modelling the stress 25mm away from the fuselage support, the approximate stresses can still be calculated, while not being affected by stress singularities.

4. The C channel is a good geometry to carry out the load in this scenario. From the total deformation seen from both shell and 3D models, it is minimal. It can also hold large amounts of normal and shear stresses without reaching the aluminium alloy's yield strength. This is because the current C channel has the maximum thickness and a tall web, increases the 2nd moment of area, ultimately making the bar stiffer and reducing deformation. The deformation plots of both shell and 3D models show convergence to a maximum value of 0.68mm. (see Figure 13)

5. The best material that should be used in an aircraft is the lighter material that can handle the loads. By analysing each available material with varying thickness, the von-mises stress and material mass was obtained.(see Figure 27) With the current thickness (5mm), the embodied energies of steel and aluminium were 249.2kg and 24.4kg. With this thickness, aluminium weighs 2.3kg while steel weighs 6.5kg. The aluminium bar thickness could be reduced to 3mm while being compliant to the 1.5 safety factor. At 3mm thickness, the mass of the bar would be 1.4kg as well. The steel bar could reach the smallest thickness of 2mm and also be compliant with the safety factor. However, at 2mm thickness, the steel bar would be 2.6kg, which is still heavier than 5mm thick aluminium. This means that aluminium is the better option for the aircraft, taking the environment into consideration, while being lightweight and surviving the loads.

6. Yes. The use of FEA shows the torsion and high stress points which would be difficult to visualise with only hand calculations. The meshes of both the shell and the 3D model tend to converge, with a small error percentage compared to the extrapolated result. This means that the structure is an appropriate to model using FEA.